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Technical & financial feasibility assessment of heat pipe evacuated tube collector for water heating using Monte Carlo technique for buildings

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A R T I C L E I N F O

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Payback period
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A B S T R A C T

The main objective of this work is to clear the economic condition of heat pipe based solar water heating technology in India. For this, a techno-economic analysis has been carried out which offers the facts for the investment in this sector. A computation model has been developed by the amalgamation of the Monte Carlo Technique (MCT) and Multi-energy/economic relations that take into consideration the risks and integrated uncertainties of different variables. The results reveal that LCWH and PP are found to be lowest for Zone-V and will require the least collector area. Since the highest solar radiation in Zone-V eventually reduces the ratio of annual energy demand to annual solar radiation. Thus Zone-V is the most favorable place for the installation of the solar water heater. The mean values of LCWH, NPV, and PP are found to be 5.14 INR/kWh, 663788.48 INR, and 5.84 years respectively. However, values of LCWH, NPV, and PP through optimization were found to be lesser by 25–33%, higher by 9–28%, and lower by 37–47% respectively.

The MCT is efficient to defeat discrepancies in the results of conventional techniques by taking into consideration the range of states according to their probability to offer more accurate forecasts.

1. Introduction

1.1. Necessity and options for solar water heating technology

In the present time, the economic progress of any country depends on the consumption of energy. Primarily, energy is generated through the burning of fossil fuels which leads to the emission of greenhouse gases into the atmosphere that is responsible for the increase in temperature of the atmosphere [1]. The contrary consequences of global warming are witnessed in across the world. It is expected that surface temperatures will increase by 1.5 °C between 2030 and 2052 if greenhouse gas emissions continue to increase at this rate [2]. The only possible way out is to replace fossil fuels with renewable energy based technologies.

Solar energy is extensively available in almost all parts of the world. India being a tropical country receives abundant solar energy throughout the year. India receives almost 5×10^3 trillion kWh of solar energy per year amounting to daily average solar energy received in the range of 4–7 kWh/m^2. This is a good indication for employing solar energy operating systems in India [3].

In a realistic scenario, the hot water demand in different sectors of India is presented in Fig. 1. It is evident that the requirement for hot water is increasing over time. Among the other sectors, the hot water demand in the residential sector is highest for the years 2010, 2013, 2017 & 2022, which is on average 80% of the total hot demand [4].

Whereas demand for hot water in the commercial/Institutional and industry sector is 13% and 6% of total demand respectively [4]. It means the residential sector is the biggest one for the prospective implementation of solar water heating systems. Thus, more emphasis should be given to the installation of solar water heating systems in the residential sector in India. Presently, this demand is fulfilled by either burning LPG gas or an electric heater [5].

Solar energy based water heating is a well-proven and widespread technology in the domestic sector that can reduce the consumption of

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Abbreviation

A_{col}	Collector area, m^2
BLC_N	bank loan cost for Nth year, INR
C_o	Capital cost, INR
C_N	annual cost of solar system, INR
CF_N	cash flow for Nth year, INR
d_r	degradation rate, %
DE_r	debt-equity ratio, %
EI_r	electricity inflation rate, %
EP	electricity price, INR
I_T	solar radiation, kWh/m^2/day
i_r	bank's interest rate, %
if_r	inflation rate, %
IEC_o	initial equity cost, INR
LCWH	Levelized cost of water heating, INR/kWh
N_L	useful life of system, year
OM_N	operating & maintenance cost of system, INR
MD_r	market discount rate, %
NPV	net present value, INR
PP	payback period, year
Q_N	energy generation in Nth year, kWh
η_{th}	thermal efficiency, %
HPT-ETCS	heat pipe technology based evacuated tube collector system

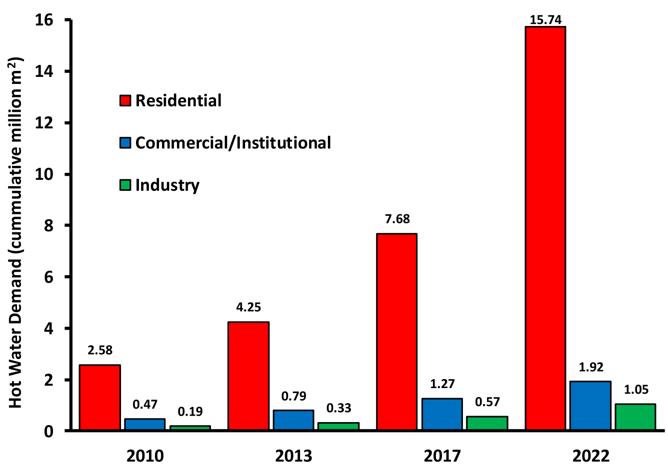


Fig. 1. Hot water demand in different sectors of India [4].

fossil fuels and the emission of carbon dioxide [6]. A lot of development in solar energy based systems have been carried out and more advanced level solar energy based water heating systems are commercializing in the market. In the present scenario, due to easy installation and higher performance, the demand for ETC (evacuated tube collector) based solar water systems are tremendously increasing. Such types of collectors offer minimal radiative & convective losses. Among different types of ETC based SWH (solar water heater), heat pipe based system is one of the most advanced and efficient ones. It may be noted that HP-ETC based SWH is suitable to supply hot water in any climate zone due to its desired characteristics such as anti-freezing properties, higher performance, and almost constant temperature level [7].

Despite a lot of benefits of utilizing heat pipe ETC, mostly thermosyphon type evacuated tube solar collector based solar water heating system has been installed for domestic and industrial sectors in India. This is because of the association of some crucial challenges with heat pipe ETC based SWH. The main challenges for the adoption of heat pipe evacuated tube based solar water heater in a country like India is its higher initial capital investment and maintenance cost due to the complexity of designing and production of heat pipes. Another major drawback is the overheating problem in heat pipes which generally occurs due to the low flow rate of heat transfer fluid or if the installed system is oversized. In addition to this, when these types of solar water heaters are utilized in domestic sectors then they have to maintain at a temperature lower than 100 °C, or else overheating problems are eventually responsible for the breaking of vacuum tubes. Furthermore, due to the high temperature of heat pipes, scaling on the condenser is also one of the major issues responsible for the blockage of the manifold of the heat pipe solar water heater.

These problems forced the negation of heat pipe based SWHs in developing countries like India. However, the authors wish to bring to notice that despite having high initial investment and maintenance costs, its high performance results in a low LCWH, high NPV, and low payback period. Besides this, overheating and scaling problems can also be overcome by using a highly efficient heat transfer or nanofluid which has a high boiling point and thermal conductivity. These solutions can enlighten the adoption of heat pipe based solar water heaters instead of thermosyphon and other designs.

1.2. Method selection for techno-economic analysis

Various studies are available concerning techno-economic calculations of different types of solar water heaters within various scenarios. In most of the studies, the thermal performance of solar water heaters has been assessed in different regions, and then evaluated the financial profits based on input economic influencing factors through various modeling techniques [8–11]. Numerous economic parameters such as LCWH (Levelized cost of water heating) [12], NPV (net present value) [13], ROI (rate of return) [14], IRR (internal rate of return) [15], UCE (unit cost of energy) [13], BCR (Benefit to cost ratio) [16] and PP (payback period) [17] have been evaluated by various economist, among these parameters, LCWH, NPV, and PP are observed to be key and most popular parameters. Other than the variety of economic factors, there is a significant discrepancy in the financial impact parameters, such as discount rate electricity cost, electricity inflation rate, inflation rate, life, bank interest, locality, etc. These parameters are responsible for a remarkable effect on the calculation of economic performance and investment decisions.

In previous literature, economic analysis of solar based water heating systems has been carried out by either self-developed models [18] or by commercialized software such as MATLAB [10], TRNSYS [19], etc. Even though these studies have appropriately used the performance parameters of the main economy, mostly used simplified models of economics to determine the performance-economic of solar water heaters. These are generally excessively optimistic due to confined series of input-assumptions and overlooking the impreciseness of each input variable [20,21]. This results in a deviation of calculated values from real cases. Therefore, the estimation of the sensitivity of the performance-economic input variable of solar water heater is not so clear and required to evolve modeling clarification to enhance the robustness of the economic valuations of system technology. The MCT is the only solution to encounter these issues. This method is a comparatively simple and proven technique for considering risk and uncertainty in a quantifiable model. In this method, estimation is carried out by selecting each input variable from Pre-set distributions for each calculation [13]. Presently, there are very few studies are available concerning to economics study of the solar water heater using MCT [13,22, 23], particularly no study has been found regarding the economic analysis of HPT-ETCS using MCT. For instance, Rout et al. [13] carried out a scenario analysis of solar water heater in the east-coastal region of India by taking three types of scenarios namely pessimistic, current and optimistic to understand the behavior of NPV (net present value). They also did the risk valuation of SWHs by using the Monte Carlo simulation model. Based on the assessment they concluded that the installation of solar collectors in the east-costal region of India is an economically viable solution. Rezvani et al. [23] performed Monte Carlo analysis of flat plate solar water heater to estimate the reliability and techno-economic profits. Also, based on the probability model, the mutable maintenance cost of a flat plate solar water heater has been determined by the authors. In addition to this, a sensitivity investigation was done to demonstrate the influence of auxiliary/supplementary heating on the economic feasibility of the collector. The investigation revealed that a flat plate solar water heater can offer remarkably long-term economic feasibility in comparison to an electricity based water heater. Inamdar and Gandhare [24] used multi-criteria and analytical hierarchy process approaches to assess four alternatives (fuel or gas, coal or biomass, solar water heater, and electricity) for water heating and also rank these alternatives. They performed this assessment based on ten identified criteria i.e. social impact, cost, technical (maintenance, availability, design, training, and safety), environment (influence on pollution & ecosystem), location plant size (flexibility & size). They found fruitful conclusions that during the selection of water heating, pollution is the least important criterion, followed by maintenance and then safety. Rout et al. [22] did an economic assessment of solar water heater using Monte Carlo based technique. In this study, operating/maintenance cost, capital cost, operation time and service life were recognized as extremely sensitive input factors. Using the Monte Carlo method, average values of exergy and mass costs of hot water were found to be 0.310 US cent/kg and 0.302 US cent/kg respectively. They suggested that the methodological method can be extended to investigate the probabilistic exergo-economic cost of a group of solar collectors when parametric uncertainties play the main role. Zhang et al. [14] provided an insight view of the different existing techniques of techno-economic for solar water heating systems. They found that the payback period of solar water heating systems is highly influenced by policy support, natural conditions, auxiliary energy type, and technology type. To conduct a techno-economic analysis of solar water heaters, a regression model was developed by them. They found that there are seven positive impacts (consumption configuration of hot water, hot water temperature, experiment, simulation, case study, investigation done in South America and Asia) and nine negative impacts (area of solar collector, evacuated tube collector, solar radiation, diesel as an auxiliary fuel, residential area, particular region, computer modeling and study done in Europe) on payback period of solar water heater.

1.3. Novelty of manuscript

One of the foremost barriers to denying the adoption of solar water heating systems in India is its high initial cost. In other words, there is an opinion among Indian people that it is uneconomical. The main aim of this study is to check the validity of this observation using Monte Carlo Analysis. In existing economic analysis techniques, the estimation of output matrices such as LCWH, NPV & PP has not taken into consideration the uncertainty as each input parameter provides merely an “optimal performance” value. This is because input parameters are influenced by many factors that can dramatically vary the value of considered input parameters.

In previous models, authors generally used the average value of input parameters, however, MCT is efficient to defeat this discrepancy by taking into consideration the range of states according to their probability. Therefore, this technique offers a more accurate forecast [25]. To date, no study particularly addresses the economic estimation of HPT-ETCS for different zones of India. Still, there are challenges to the deployment of HPT-ETCS in India, such as uncertain position/size of the solar market, different zones of India with suitable solar radiation levels, undefined policy/pricing influence & performance of economic and weak strategy of development, etc.

Therefore, the present work bridge these research gaps, by economic estimation of HPT-ETCS in India. Hence, based on the amalgamation of MCT and multi-energy/financial relations, a mathematical model has been developed that takes into consideration the risk and inherent uncertainties of different input parameters. In this model, nine critical input parameters, i.e. solar radiation, electricity price, thermal efficiency, capital cost, bank interest rate, inflation rate, O&M cost, market discount rate, and debt-to-equity ratio have been considered for the estimation of LCWH, NPV, and PP. In combination with a market study of solar water heater technology in India, it is anticipated that the present work could clear the economic condition of solar water heating technology in India and offers facts for the investment in solar water heater in the building sector.

2. Market trend of solar water heating systems

It may be possible that today’s results of the economic analysis would not be the same if the study is carried out in another nation/location or even in the same nation/location as policies, costs & tariffs are continuously varying. To make the correct evaluation, the market structure and local policies are required to be judged [26]. Presently, the Indian solar market is very large and the deployment of solar water heaters continuously increasing. As shown in Fig. 2, according to a report submitted by the Ministry of New and Renewable Energy (MNRE), in 2010, a total of 2.58 and 0.2915 million m2 capacity of solar water was installed in the residential & commercial sectors respectively. In 2013, the installation of solar water heaters almost grew by 65% and 77% in

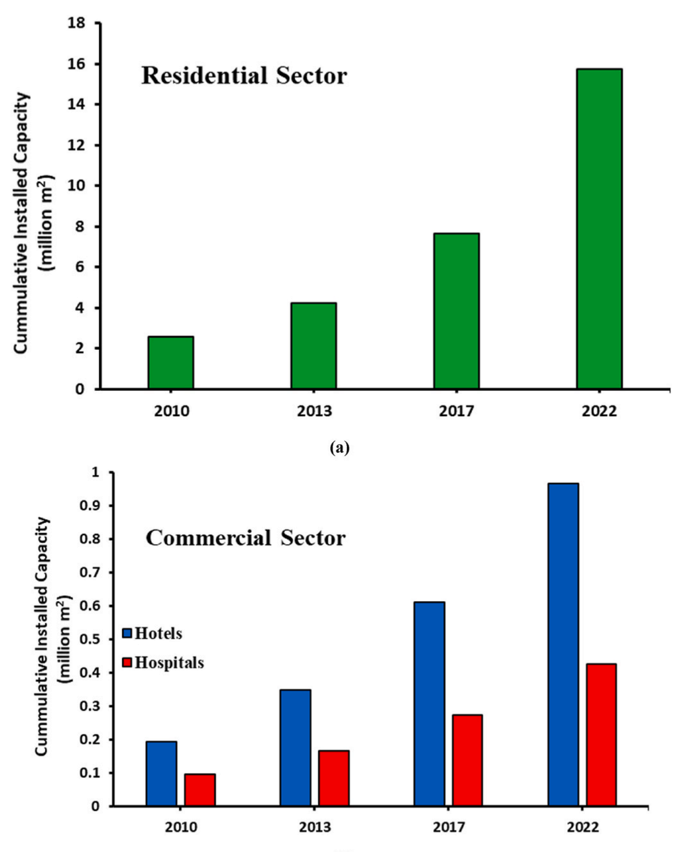


Fig. 2. Cumulative installed capacity of solar water in (a) Residential sector and (b) Commercial sector.

the residential and commercial sectors respectively. The cumulative deployment of solar water heaters in the residential sector is predicted to grow from 2.58 million m2/year in 2013 to 15.73 million m2/year in 2022. Whereas in the commercial section it is expected to grow from 0.29 million m2/year in 2013 to 1.39 million m2/year in 2022 [4].

The reason for the growth in the installation of solar water heaters between 2013 and 2022 is due to the introduction of government subsidy on initial capital cost, awareness among the local public, decreasing bank interest rates, and reforming policies by the government to simplify the rules & regulation for manufacturer [27].

According to a recent report, an ETC solar water heating system dominates the solar market over other solar water heating systems. Moreover, almost 87% of the total newly installed solar energy based water heating system that has been installed are ETC based solar water heating systems. Moreover, sales of flat plate solar collector based solar water heating systems sharply declined by 24% due to less demand. On the other hand, due to the closing of the national support scheme for the concentrated collector, their demand also got significantly lower in 2019 [28]. Hence solar market in the thermal field is shifting towards an evacuated tube collector system for low/medium temperature applications.

To attain the success of solar water heating systems in India, explicit information such as risks of investment, suggesting innovative models, the fixed suitable price of solar water heating systems, and estimated working policies of companies will be required.

3. Monte Carlo based model

The Monte Carlo analysis method has been established on Oracle Crystal Ball software. It is an application based on a spreadsheet for forecasting, optimization, simulation, and predictive modeling. This software offers an accessible and realistic technique for modeling uncertainty assisting in insight into the crucial parameters influencing risk so that the decision-makers could take a factual decision [29]. In the present model, the considered input parameters have been expressed as the triangular probability distribution, varying between maximum and minimum values (displayed in Table 1). Then these input parameters have been inserted into formulas of estimation matrics expressed in forecast cells. In this work, simulation has been repeated for 10,000 trials. These have been expressed in graphical form to analyze the reliability and sensitivity of different input parameters and to evaluate the certainty/probability of different economic output matrices [29].

The objective of the present study is to spread awareness of the cost bear for the installation of solar water heaters in a particular zone. This investigation is very beneficial to discover the financial viability of different zones of India. In this study techno-economic analysis of HPT-ETCS has been done in which the term techno specifies the technical information about components of a solar collector such as type of solar collector, model no, manufacturer, etc and the term economic specifies financial parameters such as LCWH, NPV, and PP.

Table 1
Technical specification of selected evacuated tube collector based solar water heater.

S. No	Particular	Specification/Value
1	Manufacturer	Apricus
2	Model no.	ETC-10
3	Dimensions (L × W × H)	2005 × 796 × 136 mm
4	Gross area	1.59 m^2
5	Aperature area	0.947 m^2
6	Peak output	671 W
7	Max operating pressure	8 bar
8	Gross dry weight	35 kg
9	Y-Intercept (η_0)	0.656
10	First coefficient of loss (a_1)	1.4
11	Second coefficient of loss (a_1)	0.007

The present section develops a Monte Carlo based techno-economic model for HPT-ETCS because of the present value of annual savings from the collector. A heat pipe based evacuated tube collector (ETC-10) from the Apricus Company has been taken as a reference for heat supplies in the building. In comparison to other options available for water heating applications, HPT-ETCS has numerous benefits as elaborated in the introduction section. The Monte Carlo based techno-economic model for HPT-ETCS can be illustrated in Fig. 3.

3.1. Reference HPT-ETCS

Fig. 4 illustrates the basic configuration of reference HPT-ETC technology. The heat pipe evacuated tube collector system is comprised of main four components heat pipe, evacuated tube, manifold & mounting frame. The solar radiation incident on the evacuated tube converted into useful thermal energy. The vacuum across the outer and inner glass tubes insulates against convection/radiation losses. The fins around the heat pipe assist in transferring the absorbed heat by the absorber tube to the evaporator of a heat pipe. Further, heat is transferred from the evaporator to the condenser of a heat pipe.

Finally, heat is transferred to flowing water in an insulated manifold through the condenser. The mounting frame is used to support the different components of the collector. In this study, ETC-10 with dimension (L × W × H) of 2005 × 796× 136 mm has been used as a reference collector whose aperture and gross areas are 0.947 m^2 and 1.59 m^2.

Based on the bulletin released by the manufacturer, the thermal efficiency of the collector is the function of inlet water temperature, ambient temperature, and solar radiation, which can be expressed as $\eta = \eta_o - 1.4\,(T_m - T_{amb}/I_T) - 0.007 I_T (T_m - T_{amb}/I_T)^2$. The technical specification of reference HPT-ETCS is given in Table 1.

3.2. Energy & exergy production model

The model for thermal energy production is the function of daily average solar radiation, thermal efficiency, collector area, and rate of degradation. The model of energy generation can be defined through the following equations [28].

$$Q_N = (365 \times I_T \times \eta_{th} \times A_{col} \times \Delta\tau) \times (1 - d_r)^N \tag{1}$$

3.3. Economic estimation matrices

The Levelized cost of water heating (LCWH) measures the unit costs of working a system throughout its whole life. All financial elements like debt-equity ratio, electricity price, inflation rate, initial capital cost, bank interest rate, operating and maintenance cost, rate of degradation, and discount rate are taken into consideration while estimating LCWH. The equation for the calculation of LCWH can be expressed as:

$$LCWH = \frac{\sum_{N=0}^{N} \frac{C_N}{(1+MD_r)^N}}{\sum_{N=0}^{N} \frac{Q_N}{(1+MD_r)^N}} \tag{2}$$

While the total yearly cost of the thermal collector could be given as:

$$C_N = (IEC_o \,|\, N = 0) + BLC_N + OM_N \tag{3}$$

The initial capital investment (ICC_o) on the thermal collector could be defined through the following equation:

$$IEC_o = C_o \times (1 - DE_r) \tag{4}$$

whereas, bank loan installment for an N^{th} year could be defined as expressed through the following equations:

$$BLC_N = C_o \times DE_r \times \left(\frac{1}{N_L} + \left(\frac{N_L - N}{N_L}\right) \times i_r\right) \tag{5}$$

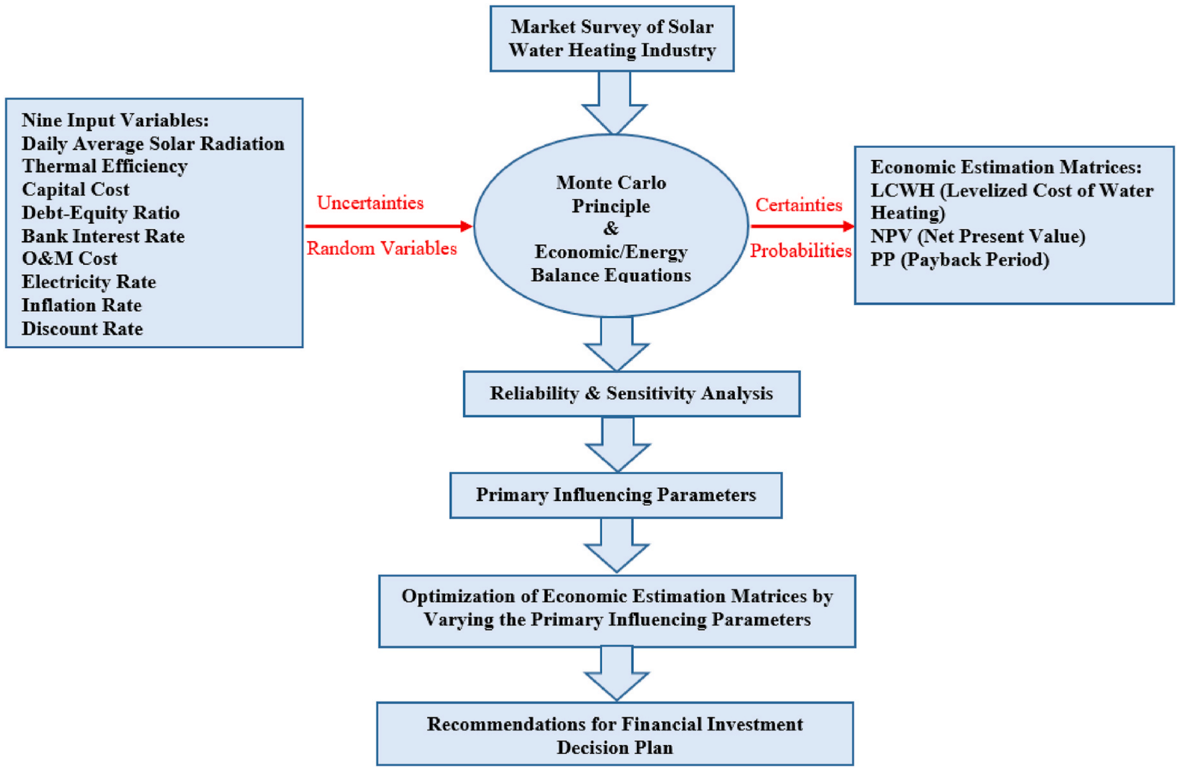


Fig. 3. Monte Caro based techno-economic model.

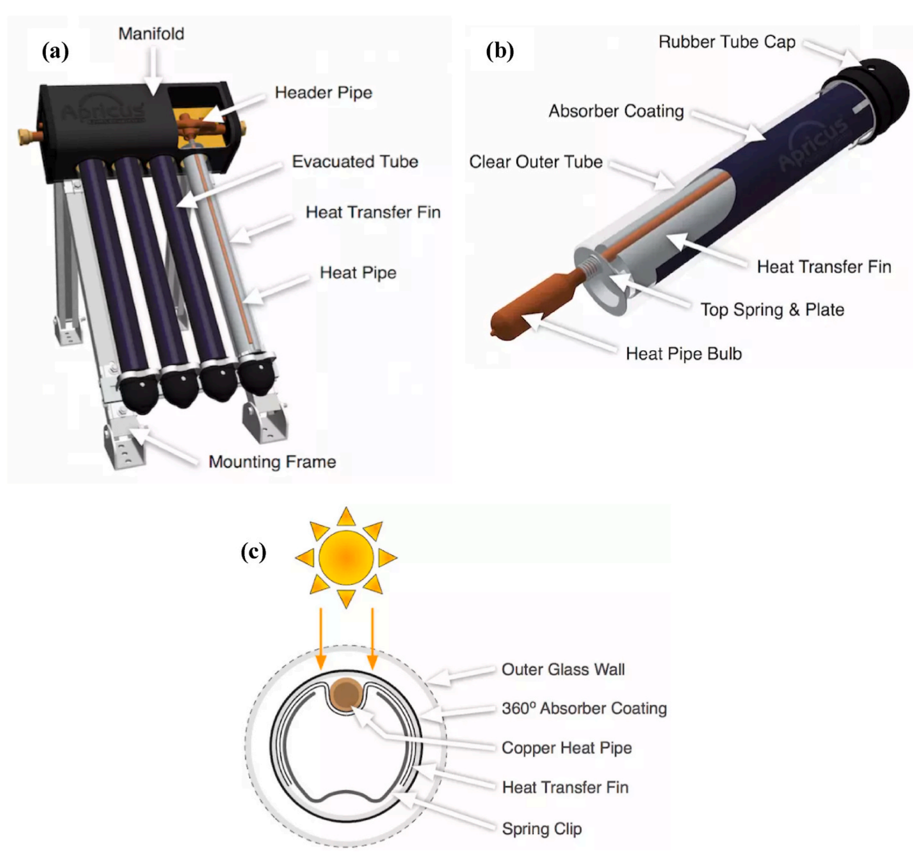


Fig. 4. (a) Heat pipe evacuated tube collector system (b) Assembly of heat pipe with evacuated tube (c) cross-section of heat pipe with evacuated tube.

The NPV is a crucial factor of economic analysis that weighs the calculation of cumulative revenue evaluated by cash inflow from cash outflow over the life of the product. It helps in the prediction of the feasibility of investment in a business project. For investment in the project, the NPV value with a positive value will be acceptable else an investment with its negative value is rejected.

$$\mathrm{NPV} = \sum_{N=0}^{N} \frac{\mathrm{CF_N}}{(1+\mathrm{MD_r})^{N}} - \mathrm{C_o} \tag{6}$$

The payback period (PP) is the time required to recover the initial capital investment in a venture. This matric of economic analysis is used to compute the economic viability of the initial capital investment. If the payback period of investment is larger than the life span of the project, then installation of the project is not admissible. In this study, if the payback period is higher than the life of the collector then it is equal to 15 years.

3.4. Vital input elements

The vital input elements to investigate the solar water heater are subdivided into metrological, technical, & financial ones. Table 1 demonstrated the basic input parameters that have been used in the simulation model, wherein the minimum & maximum values of each considered parameter have been calculated with applicable possibilities. For illustration, the daily average solar radiation of India has been considered to fit with the incessant curve of probability. In the present study, metrological data of five different climatic zones have been gathered to cover all weather conditions of India and simulation has been done based on the average of different zones. The daily average solar insolation fluctuates between 3.50 and 6.98 kWh/m^2/day and its yearly average value is 5.24 kWh/m^2/day. It can be then expressed to abide by the continuous probability distribution oscillating amongst maximum & minimum values (presented in Table 1). Therefore, it is probable to further run the probabilistic investigation by MCT, directing various combinations that abide by the probability graph of various input variables. Based on the India market survey, other input parameters such as electricity price, thermal efficiency capital cost, bank interest rate, inflation rate, O&M cost, market discount rate, and debt-equity ratio vary between 6 and 7 INR/kWh, 51–69%, 30,000–35000 INR/m^2-Aperture area, 8–10%, 4–5%, 1–1.5%, 4.5–5.5% and 50–90% respectively and their average values abide the continuous probability distribution oscillating amongst maximum & minimum values (presented in Table 2). The total estimation approach in the MCT is defined by input-output vectors in Equation (7).

4. Results and discussion

In the current study, five cities located in five different climatic zones (shown in Fig. 5) have been selected for a techno-economic estimation of a heat pipe solar water heating system to clear the economic condition of solar water heating technology. A computation model has been developed by the amalgamation of the Monte Carlo Technique (MCT) & Multi-energy/economic relations that take into consideration the risks and integrated uncertainties of different variables. The simulation has been done based on gathered metrological data from these five cities. Table 3 presents the detail of selected cities located in different zones of India. In subsequent sub-sections monthly variations of metrological data then sensitivity/reliability analysis and finally optimization analysis have been well presented and discussed.

4.1. Variation of metrological parameters

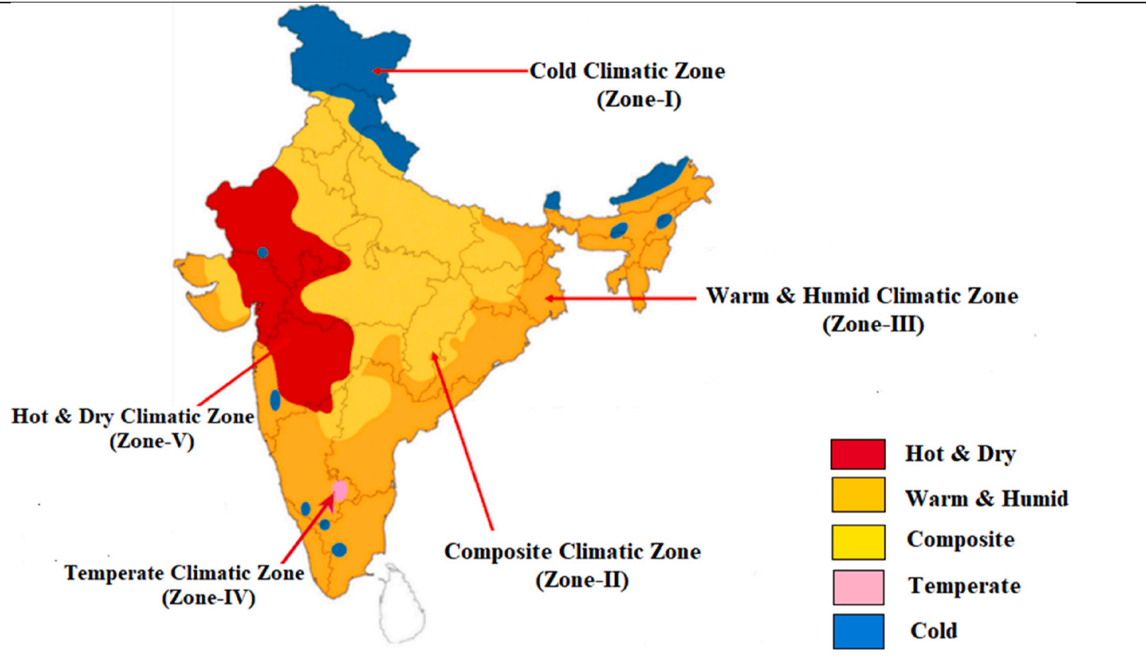
The metrological data such as solar irradiance and ambient temperature of selected cities have been gathered from the NASA database for the techno-economic analysis of the solar water heater. The monthly fluctuation of solar radiation and ambient temperature in Srinagar, Jaipur, Chennai, and Bengaluru is depicted in Fig. 6(a), (b), (c), (d), and (e) respectively. These metrological data are very crucial for the energy modeling of the solar water heater. The global horizontal irradiance in Zone-I, Zone-II, Zone-III, Zone-IV & Zone-V was found to be in the range of 2.71–6.73 kWh/m^2/day, 3.79–7.08 kWh/m^2/day, 3.95–6.50 kWh/m^2/day, 4.40–6.53 kWh/m^2/day and 4.21–7.55 kWh/m^2/day respectively.

Whereas, fluctuation of solar radiation on the tilted collector (at respective latitude) in Zone-I, Zone-II, Zone-III, Zone-IV & Zone-V has been observed in the range of 4.27–6.98 kWh/m^2/day, 4.30–6.72 kWh/m^2/day, 3.88–6.68 kWh/m^2/day, 4.48–6.71 kWh/m^2/day and 3.54–7.12 kWh/m^2/day respectively. Among the selected cities, on average, Ahmedabad and Srinagar achieve the highest (5.615 kWh/m^2/day) and lowest (4.695 kWh/m^2/day) daily average solar radiation respectively. The yearly mean value of ambient temperature in Zone-I, Zone-II, Zone-III, Zone-IV & Zone-V varied in the range of –5.93 °C (in January) to 19.04 °C (in August), 13.10 °C (in January) to 34.13 °C (in June), 24.59 °C (in December) to 31.18 °C (in May), 19.74 °C (in December) to 28.43 °C (in May) and 17.49 °C (in January) to 35.71 °C (in May) respectively. This signifies that Zone-I and Zone-III are the

Table 2
Key Input Parameters for the simulation model.

$$\begin{bmatrix} \mathrm{LCWH}^1 & \mathrm{NPV}^1 & \mathrm{PP}^1 \\ \mathrm{LCWH}^2 & \mathrm{NPV}^2 & \mathrm{PP}^2 \\ \mathrm{LCWH}^3 & \mathrm{NPV}^3 & \mathrm{PP}^3 \\ \vdots & \vdots & \vdots \\ \mathrm{LCWH}^N & \mathrm{NPV}^N & \mathrm{PP}^N \end{bmatrix} = \begin{bmatrix} \mathrm{I_T^1} & \mathrm{EP^1} & \eta_{th}^1 & \mathrm{C_0^1} & \mathrm{i_r^1} & \mathrm{if_r^1} & \mathrm{OM_c^1} & \mathrm{MD_r^1} & \mathrm{DE_r^1} \\ \mathrm{I_T^2} & \mathrm{EP^2} & \eta_{th}^2 & \mathrm{C_0^2} & \mathrm{i_r^2} & \mathrm{if_r^2} & \mathrm{OM_c^2} & \mathrm{MD_r^2} & \mathrm{DE_r^2} \\ \mathrm{I_T^3} & \mathrm{EP^3} & \eta_{th}^3 & \mathrm{C_0^3} & \mathrm{i_r^3} & \mathrm{if_r^3} & \mathrm{OM_c^3} & \mathrm{MD_r^3} & \mathrm{DE_r^3} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \mathrm{I_T^N} & \mathrm{EP^N} & \eta_{th}^N & \mathrm{C_0^N} & \mathrm{i_r^N} & \mathrm{if_r^N} & \mathrm{OM_c^N} & \mathrm{MD_r^N} & \mathrm{DE_r^N} \end{bmatrix} \tag{7}$$

Parameter	Minimum	Maximum	Average	Unit
Metrological Parameters				
Average solar radiation	3.50	6.98	5.24	kWh/m^2/day
Electricity price	6	7	6.50	INR/kWh
Technical Parameters				
Aperture area of collector	–	–	6.32	m^2/collector
No. of Evacuated Tubes	–	–	67	–
Operational period	–	–	15	year
Average thermal efficiency	51	69	60	%
Seasonal efficiency of electricity	–	–	90	%
Degradation rate	–	–	1	%
Financial Parameters				
Initial capital cost (Include direct & indirect cost)	30,000	35,000	32,500	INR/m^2-Aperture area
Bank interest rate	8	10	9	%
Inflation rate	4	5	4.5	%
O&M cost	1	1.5	1.25	%
Market discount rate	4.5	5.5	5	%
Loan period	–	–	15	year
Debt to equity ratio	50	90	70	%
Electricity inflation rate	–	–	10	%
Other Parameters				
Average number of people in a House	–	–	6	–
Daily Hot Water Requirement	–	–	60	L/day/Person
Occupancy rate	–	–	100	%
Desired hot water temperature	–	–	60	°C
Solar tracking mode	–	–	Fixed	–
Azimuth	–	–	0°	–
Inclination angle of Collector	–	–	Latitude	–
Domestic solar water heating technology	–	–	ETC	–
Traditional fuel type	–	–	Electricity	–



Climatic Zone	Srinagar (Zone-I)	Jaipur (Zone-II)	Chennai (Zone-III)	Bengaluru (Zone-IV)	Ahmedabad (Zone-V)
Solar Radiation (kWh/m2/day)	2.71-6.68	3.79-7.08	3.95-6.50	4.40-6.45	3.68-7.55

Fig. 5. Different climatic zones of India.

Table 3
Detail of selected cities falls in different climatic zones of India.

Zone	Climatic Zone Type	Location (Longitude, Latitude	Average Relative Humidity (%)	Average Precipitation (mm)	Elevation (m)
Zone-I	Cold (Sunny/ Cloudy)	Srinagar (74.80° E, 34.1° N)	70.4	651.56	1593
Zone-II	Composite	Jaipur (75.78° E, 26.91° N)	50.0	497.42	390
Zone-III	Warm & Humid	Chennai (80.27° E, 13.08° N)	73.6	1213.94	25
Zone-IV	Temperate/ Moderate	Bengaluru (77.59° E, 12.97° N)	68.6	852.4	919
Zone-V	Hot & Dry	Ahmedabad (72.57° E, 23.02° N)	56.7	689.98	55

coldest and hottest zones among the selected zones.

4.2. Sensitivity/reliability investigation

In this section, a sensitivity investigation has been carried out to gauge the sensitivity of the fluctuation in vital input parameters for the estimation of output matrices (LCWH, NPV & PP). In this investigation impact of considered input parameters on the output matrices (LCWH, NPV, and PP) has been analyzed (as depicted in Fig. 7).

As shown in Fig. 7 (a), an average value of LCWH is found to be at 5.14 INR/kWh. At the highest frequency, there is approximately a 6% probability that the value of LCWH is found to be around 5 INR/kWh. Moreover, there is nearly 90.65% certainty that the value of LCWH swings between 3.80 and 6.00 INR/kWh over the life (15 years) of HTP-ETCS (heat pipe evacuated tube collector system). However, the average value of LCWH by conventional energy source (electricity) is falling in the range of 6.50–28.05 INR/kWh for the said period. This is because HTP-ETCS bear lower annual costs to accomplish annual energy demand than conventional energy based systems. On the other hand, the average value of NPV of HTP-ETCS was observed to be 663788.48 INR after 15 years of working (as depicted in Fig. 7 (b)).

It is more interesting to note that there is 100% certainty to accomplish the positive value of NPV when it is worked in India. Under the judgment rule of NPV, a value of NPV greater than zero specifies that investment in the HPT-ETCS will be a profitable one as the estimated profit surpasses the predicted costs.

As depicted in Fig. 7 (c), the mean value of the payback period is approximately 5.84 years when the calculated cash flow turns positive. Also, there is 100% certainty for HPT-ETCT to have a payback period of less than 15 years. Also at the highest frequency, there 8.5% probability to have a payback period of 6 years.

The sensitivity analysis is very easier to discover which input variable affects the most the forecasted output and decreases the time required to improve estimates. In another way, the input variables whose influence is least on the forecast can be discarded or ignored together. Consequently, a more realistic model can be constructed that significantly enhance the preciseness of the results.

The sensitivity of considered input variables on the output parameter is depicted in Figs. 8–10. The key variables that govern the positive results of commercial/business cases concerning heat pipe evacuated tube solar collector systems are average solar radiation, electricity price, average thermal efficiency, initial capital cost, nominal interest rate, inflation rate, O&M cost, market discount rate, debt to equity ratio and electricity inflation rate. As depicted in Fig. 8, the average solar insolation assumption has the utmost impact on the uncertainty of LCWH by 79.47%, followed by the thermal efficiency assumption (15.65%). The capital cost has also marginally influenced the uncertainty of the LCWH

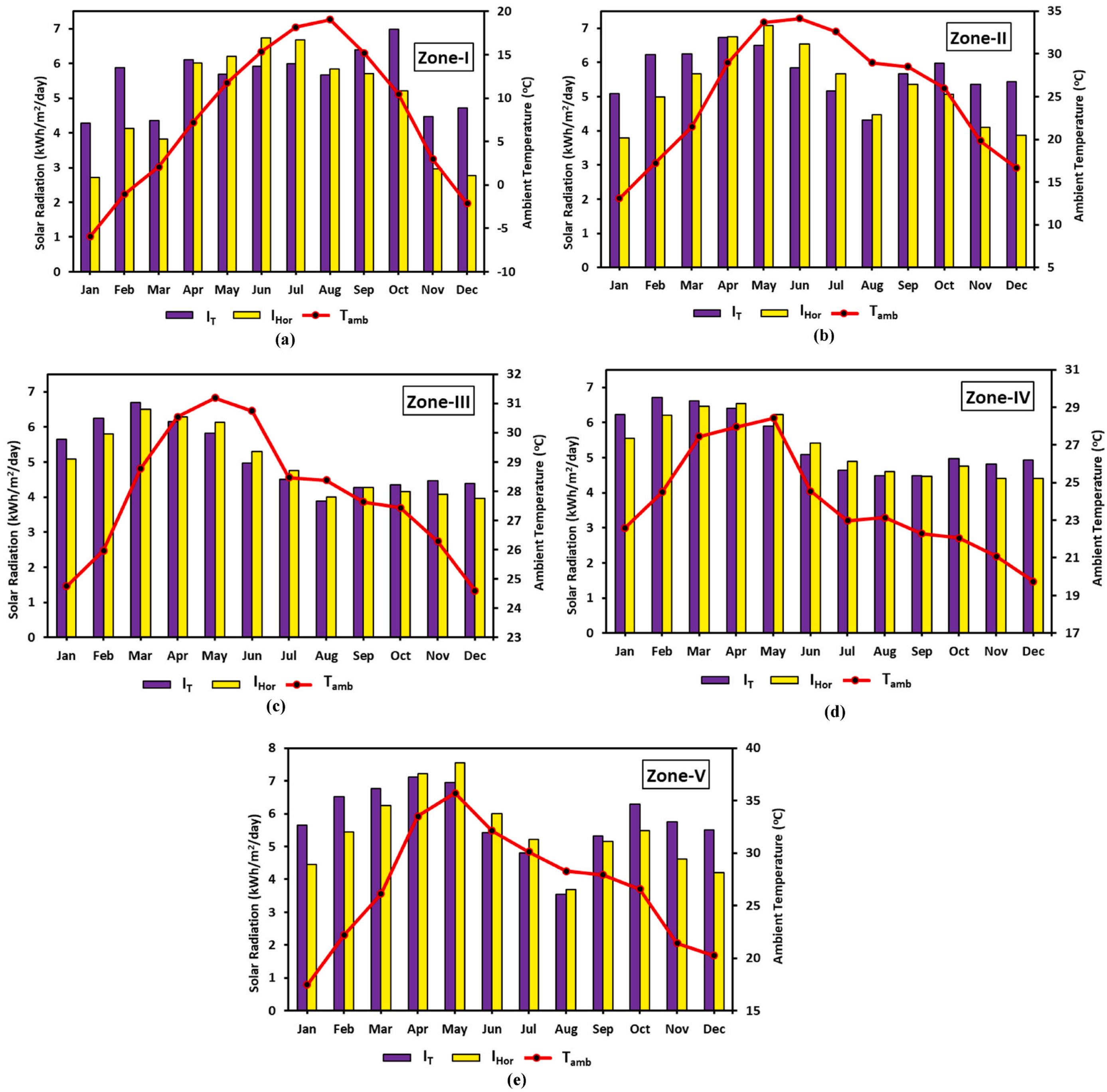


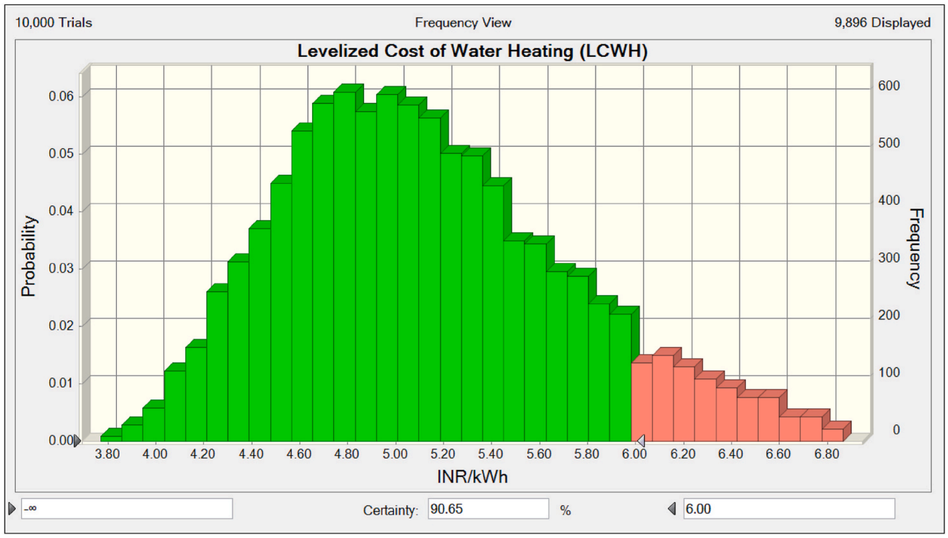
Fig. 6. Monthly fluctuation of solar irradiance on tilted/horizontal surface and ambient temperature for (a) Zone-I (b) Zone-II (c) Zone-III (d) Zone-IV and (e) Zone-V.

by 3.05%. This means that before offering the predicted value of LCWH in front of buyers, the company's priority is to take the daily average solar radiation data of the particular location.

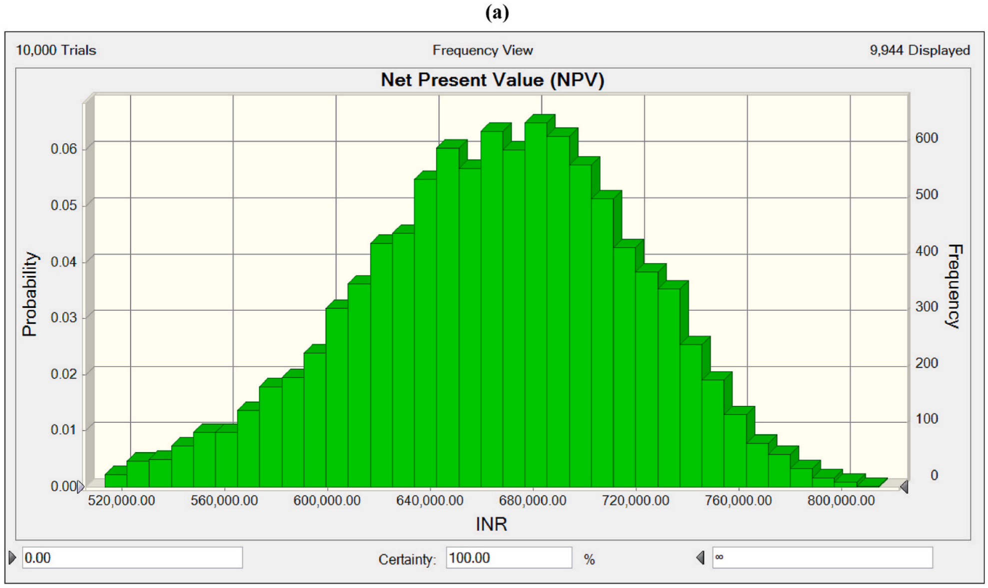
As it governs the generation of hot water production and yearly savings of HPT-ETCS. The capital cost and thermal efficiency have less degree of influence on the uncertainty of LCWH. Since the total yearly cost of the solar system is greatly influenced by capital cost and thermal efficiency. However, the rest of the assumptions have the least impact on the uncertainty of LCWH.

Similarly, the uncertainty of NPV (as depicted in Fig. 9) is intensely impacted by the average daily solar radiation by 50.31%. The uncertainty of NPV is also significantly sensitive to electricity price, thermal efficiency, discount rate, and capital cost by 31.10%, 9.48%, 5.82%, and 2.17% respectively. As NPV is the function of discount rate and cash flow and these components directly contribute to the cash flow and yearly savings. Therefore, these components based on priority should be considered before calculating NPV. The rest of the components have a very mere impact on the uncertainty of NPV.

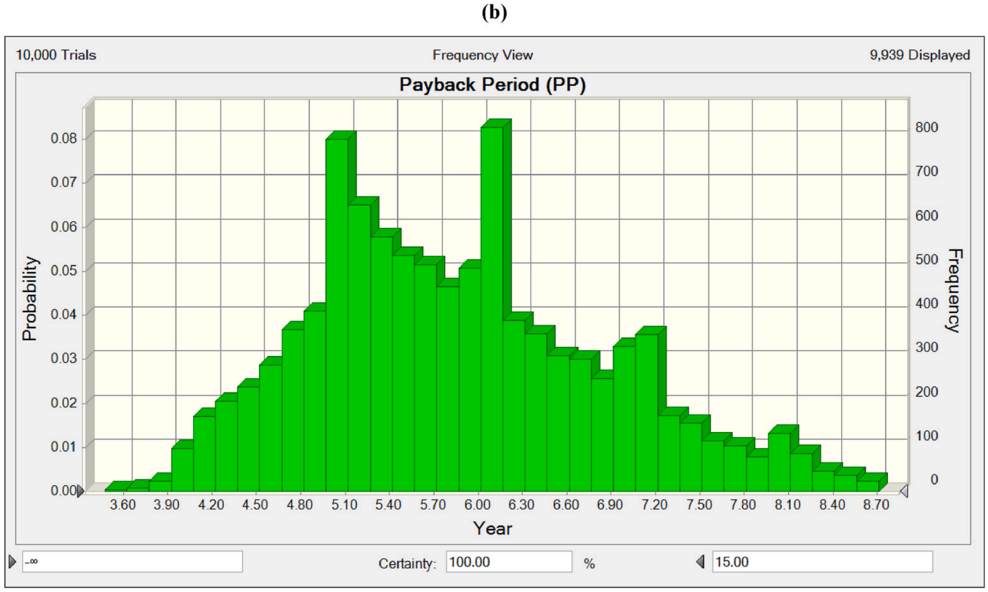
The payback period is the period of recovery of initial investment on any business project. This wholly depends on the profit gained by installing HPT-ETC. The profit from the solar system is directly connected to the amount of energy generation (which directly depends on the incident of average daily solar radiation), followed by thermal efficiency, debt-equity ratio, electricity price, and capital cost. As shown in Fig. 10, the remaining components are not so important and their influence on the PP is so limited and can be discarded.



(a)

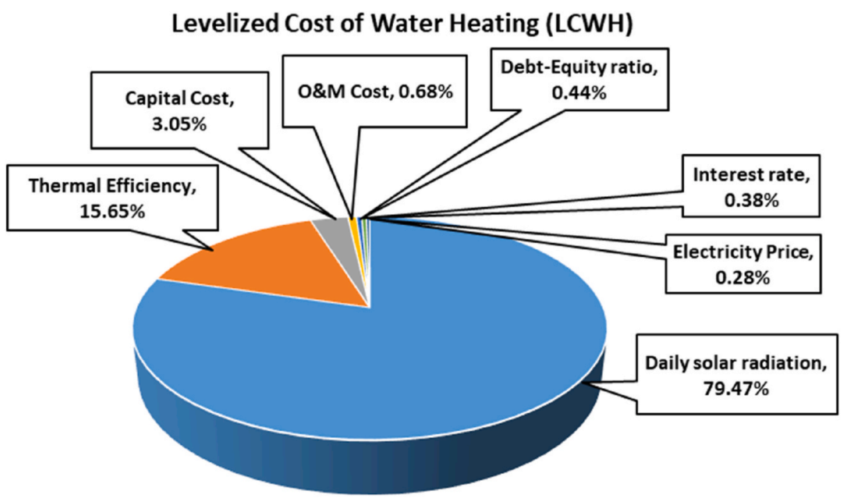


(b)

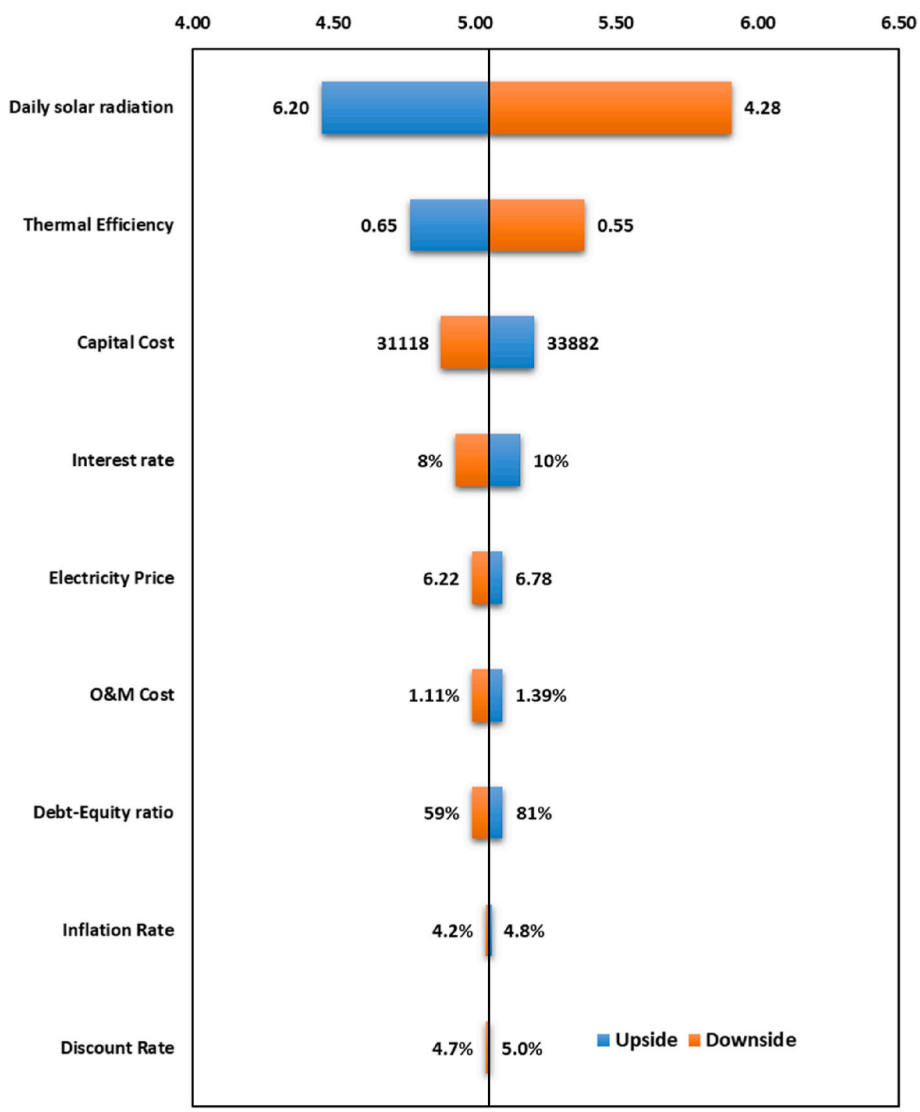


(c)

Fig. 7. Frequency and Probability forecast chart of (a) Levelized cost of water heating (b) Net present value (c) Payback period.

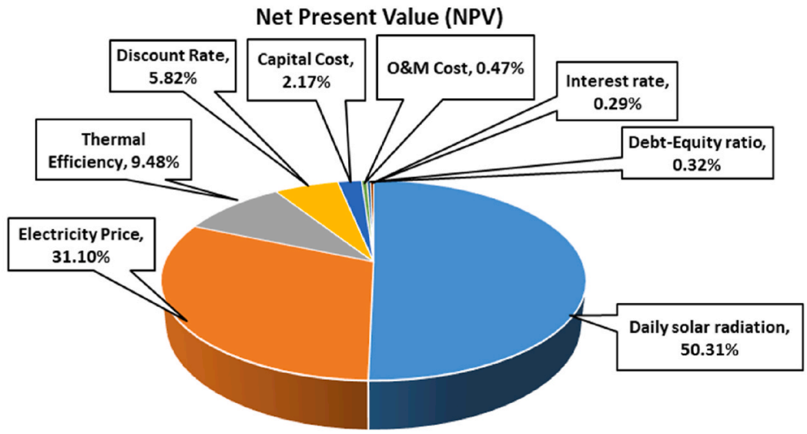


(a)

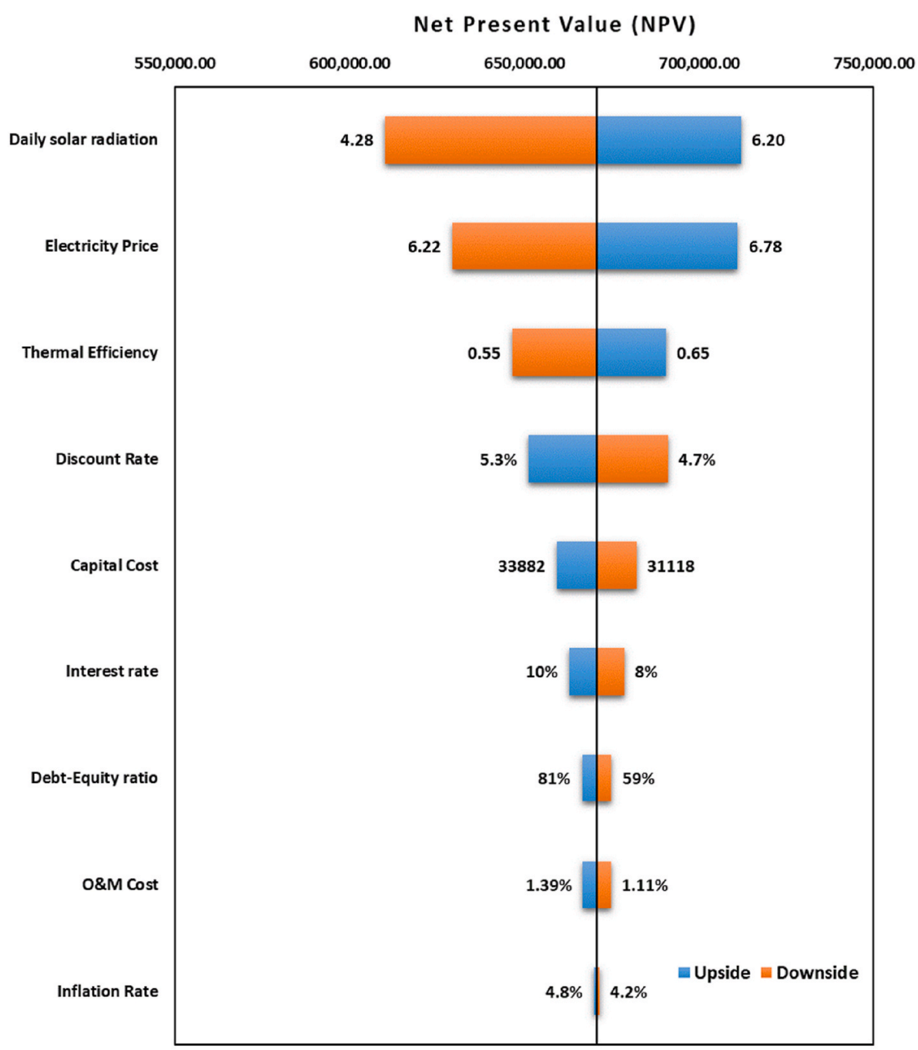


(b)

Fig. 8. (a) Sensitivity and (b) Tornado chart of LCWH.



(a)



(b)

Fig. 9. (a) Sensitivity and (b) Tornado chart of NPV.

4.3. Optimization analysis

In this section, to measure the potential influence of each input parameter on economic output matrices (LCWH, NPV, & PP), the variation of each variable has been put in a definite range while keeping other assumptions the same. The outcomes of the simulation will be beneficial in the forecasting of future solar market position and penetration of the HPT-ETCS. The discussion of this section has been

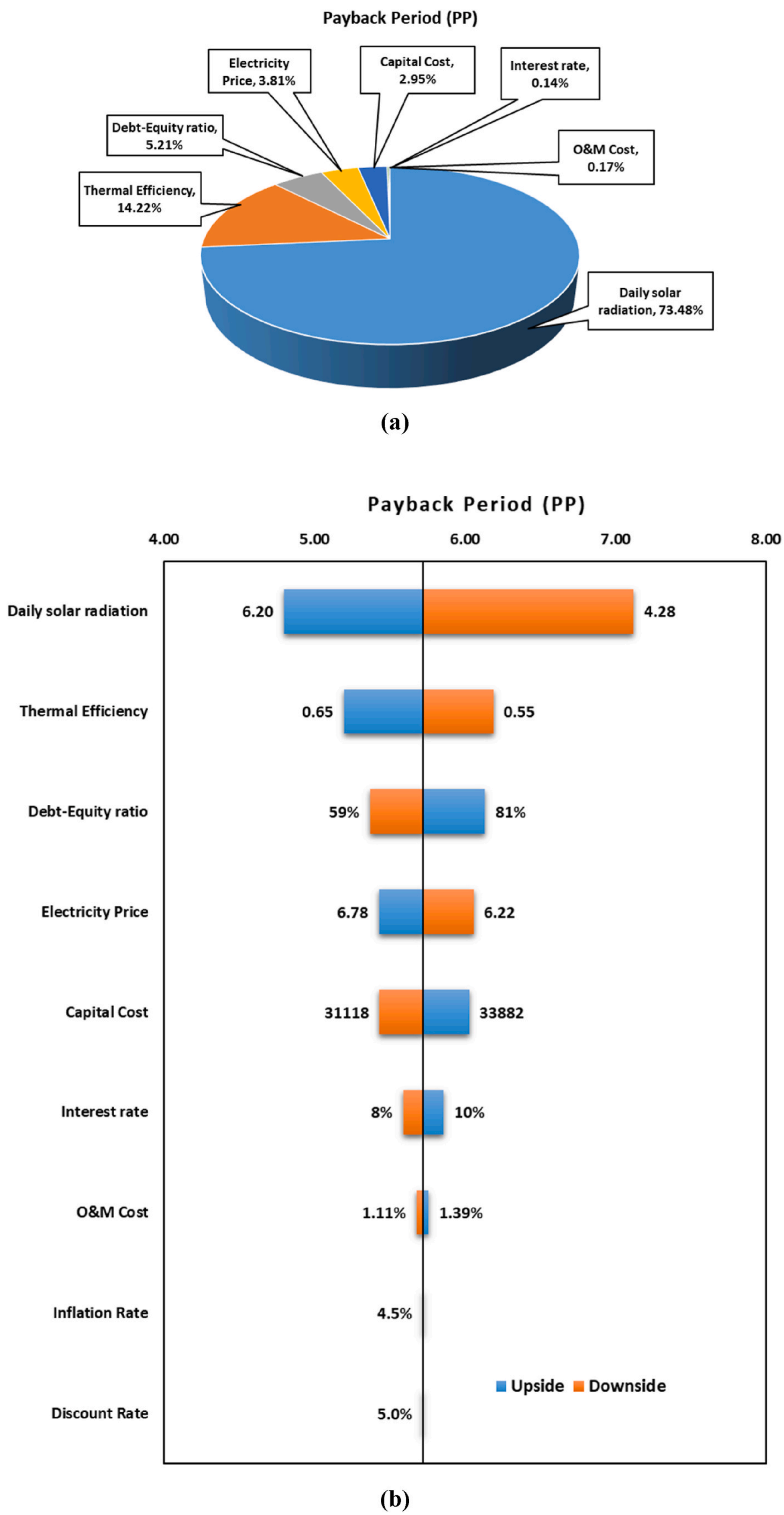


Fig. 10. (a) Sensitivity and (b) Tornado chart of PP.

followed by the optimization of LCWH, NPV, and PP using the genetic algorithm method.

4.3.1. Effect of daily average solar radiation

The impact of daily average solar radiation on LCWH, NPV, and PP has been depicted in Fig. 11 when only average daily solar radiation increased from 1 to 7 kWh/m^2/day by keeping other assumptions the same in the designed model. It may be noted that only daily average solar radiation has been supposed as a static value for each simulation and the other variables follow triangular distributions.

Fig. 3 shows that NPV increases, whereas both LCWH and PP decrease with an increase in average daily solar radiation. It has been observed that NPV increases from −462546.33 to 737880.25 INR ($y = 1833.2x^3 - 50497x^2 + 468783x - 815246$, $R^2 = 0.9871$) in a polynomial way. It is also observed that NPV turns positive when the average daily solar radiation becomes higher than approximately 1 kWh/m^2/day.

LCWH was found to decrease from 21.17 to 4.09 INR/kWh ($y = 22.346x^{-0.657}$, $R^2 = 0.998$). Similar to the NPV, the PP in a polynomial way ($y = -0.0028x^4 + 0.082x^3 - 0.7508x^2 + 1.122x + 14.79$, $R^2 = 0.9963$) decreased from 16.97 years to 4.13 years.

As shown in the figure, the payback period of the collector would be less than 15 years when the daily average solar insolation will be higher than 1.5 kWh/m^2/day. It will be the minimum value of 4.13 years when the mean daily solar irradiation reached almost 6.98 kWh/m^2/day. Results inferred that higher solar radiation results in more generation of energy & surges the yearly savings and NPV. Furthermore, higher solar radiation has a positive influence on economic parameters and provides better results for the business.

Hence, heat pipe evacuated tube collectors are suggested to be installed where at least average daily solar radiation is higher than 2 kWh/m^2/day to attain positive NPV and low value of LCWH and a payback period of less than 15 years.

4.3.2. Effect of capital price

Fig. 12 has been illustrated the impact of variation of capital price when increasing the capital price of the collector from 25,000 to 41,000 and keeping the other assumption the same in the proposed model. In this case, only capital price has been supposed as the static value for each simulation in place of triangular distribution.

This figure shows that variation in capital cost has a significant impact on LCWH, NPV, and PP. LCWH increases linearly from 4.17 to 6.04 INR/kWh ($y = 0.234x + 3.9344$, $R^2 = 1$), NPV decreases linearly from 732508.39 to 600886.53 ($y = -16453x + 748961$, $R^2 = 1$), whereas the payback period increases polynomial from 4.27 to 7.18 years ($y = -0.0035x^2 + 0.3994x + 3.9049$, $R^2 = 0.9975$). The higher capital cost of the collector results in a higher loan amount; For this reason, it has a diverse impact on economic output parameters. The simulation outcomes show that the capital cost of the collector is an important factor for the decision to invest in the solar water heater and it also advised that capital cost must be regulated as possible for better penetration of the market.

4.3.3. Effect of discount rate

Varying the discount rate & keeping other conventions the same, Fig. 13 depicts the variation trend of LCWH, NPV & PP under this condition. The discount rate has been regarded as a static value in place of triangular distribution during each simulation.

It has a large influence on the NPV, limited impact on LCWH, and no impact on PP. With an increase in the discount rate from 4% to 10%, NPV reduces almost linearly from 741259.3 to 417440.19 ($y = -26805x + 750357$, $R^2 = 0.9913$); while LCWH increases linearly from 5.02 to 5.20 INR/kWh ($y = 0.0003x^2 + 0.0097x + 5.0122$, $R^2 = 0.9972$).

It has been observed that a higher discount rate abates the energy generation factor and cash flow and it finally decreases NPV & increases LCWH. It can also be seen that; a lower value discount rate is highly appreciable for the initial investment in the HPT-ETCS.

4.3.4. Effect of debt-equity ratio

The fluctuation of LCWH, NPV, & PP with reference to the increase in debt-equity ratio has been demonstrated in Fig. 14 when the other input variables are continued as the same conventions. The D-E ratio has been regarded as the static value in the proposed model in place of triangular distribution during each simulation. It may also be noted that the effect of the D-E ratio is not as much as compared to daily average solar radiation. Both LCWH and PP increase with an increase in the D–E ratio, while the NPV decreases. As shown in Fig. 14, the variation of D-E ratio from 10% to 100% put a limited impact on LCWH and NPV. The LCWH increases from 4.72 to 5.21 INR/kWh in a linear fashion ($y = 0.054x + 4.668$, $R^2 = 0.997$), whereas NPV declines from 693504.04 to 659463.87 INR in a perfectly linear way (697286, $R^2 = 1$). On the other

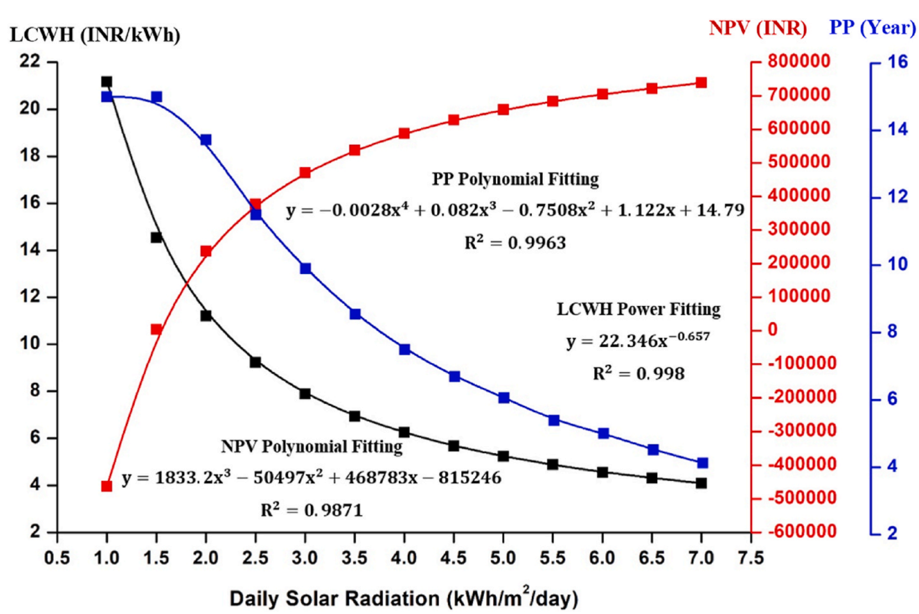


Fig. 11. Impact of variation of daily average solar radiation on LCWH, NPV, and PP.

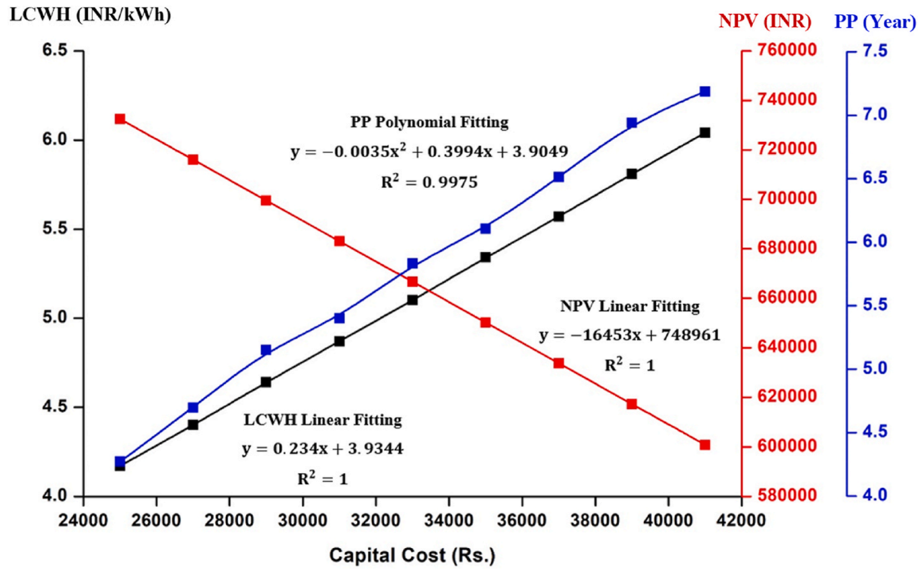


Fig. 12. Impact of variation of capital cost on LCWH, NPV, and PP.

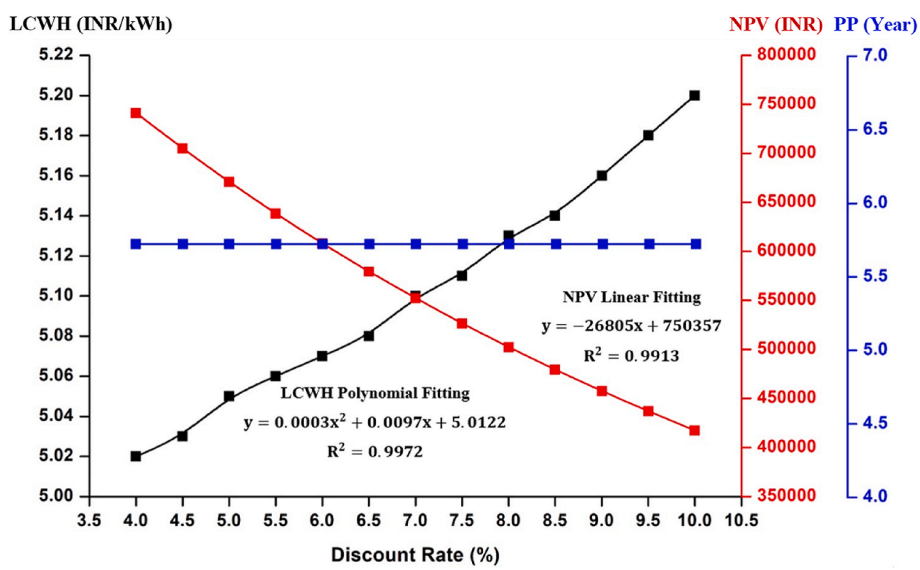


Fig. 13. Impact of variation of discount rate on LCWH, NPV, and PP.

hand, PP exponentially goes up with increasing in D-E ratio from 4.19 to 6.63 years ($y = 4.0037e^{0.05y=-3782.2x+06x}, R^2 = 0.9975$).

Because, a higher D-E ratio means a lowering of down payment, and higher loan cost, which results in higher LCWH, PP, and lower NPV due to a negative impact on cash flow. These results significantly boost Indian clients to have more credits from the bank if they want to invest in solar water heating systems which is beneficial for the expansion of advanced business models among various stakeholders.

4.3.5. Effect of electricity price

When varying the electricity price only, its effect on the LCWH, NPV, and PP has been depicted in Fig. 15. It requires to be noted that the electricity price has been regarded at a fixed value in place of triangular distribution while other input variables abide by triangular distributions. It can be seen that NPV and PP are highly sensitive to variations in electricity price while there is a limited effect on LCWH. By variation of electricity price from 4 to 8 INR/kWh, LCWH linearly ($y = 0.0963x + 4.4672, R^2 = 0.9999$) increases from 4.56 to 5.33 INR/kWh, while NPV greatly increases from 308976.65 to 886337.13 INR in a perfectly linear fashion ($y = 72128x + 237368, R^2 = 1$), on the contrary, the payback period significantly polynomial decreases ($y = 0.043x^2 - 0.9925x + 10.076, R^2 = 0.999$) from 9.18 to 4.56 years.

This is because higher electricity price leads to increases the cash flow and annual savings which consequently increases the NPV and decreases the payback period. In contrast, with an increase in electricity price, there is no impact on the generation of electricity but the total annual cost of the solar system increases that's why LCWH increases. The results showed that the installation of solar water heaters should be prioritized in those municipalities or regions of India where electricity prices are high.

4.3.6. Effect of thermal efficiency

Considering the variation of thermal efficiency only & keeping other conventions the same, the fluctuation of LCWH, NPV & PP with respect to the increase of thermal efficiency has been depicted in Fig. 16. The thermal efficiency has been set as a static value (in place of triangular distribution) during each simulation. It was found that the increase in

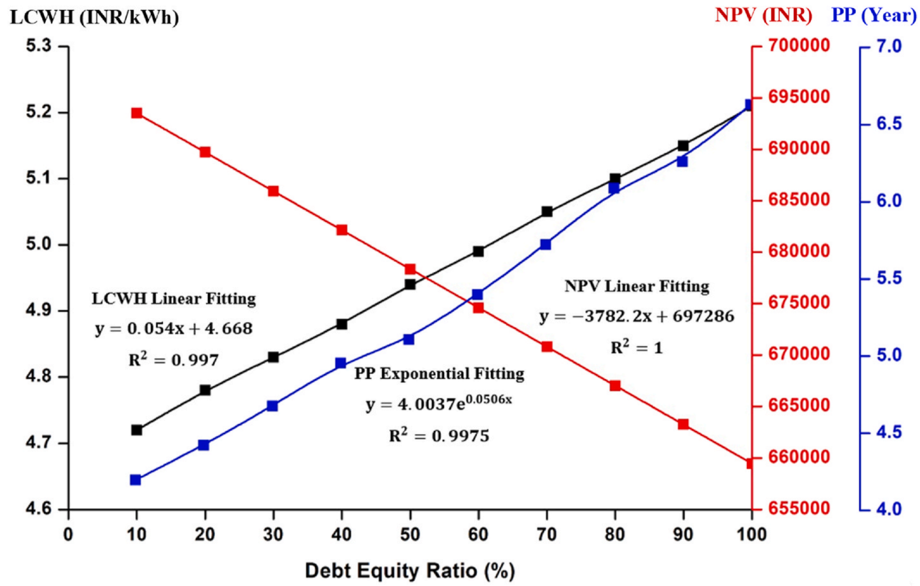


Fig. 14. Impact of variation of D-E ratio on LCWH, NPV, and PP.

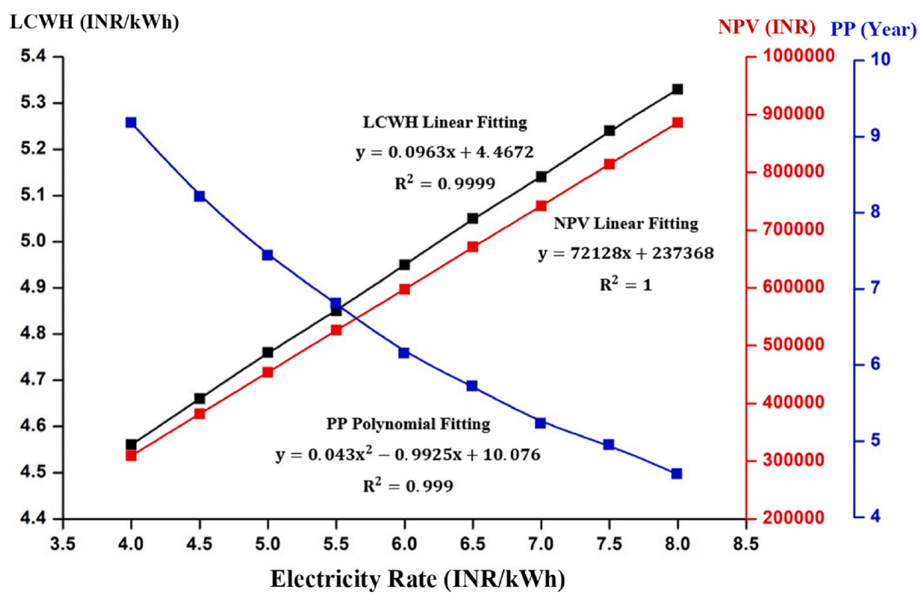


Fig. 15. Impact of variation of electricity rate on LCWH, NPV, and PP.

thermal efficiency has a positive impact on LCWH, NPV, and PP. LCWH linearly ($y = 0.0047x^2 - 0.212x + 6.5142, R^2 = 0.9998$) decreases from 6.32 to 4.38 INR/kWh, NPV increases polynomial from 581231.54 to 717796.77 INR ($y = -332.62x^2 + 14923x + 567887, R^2 = 0.9997$), while the payback period decreases in a polynomial way ($y = 0.0069x^2 - 0.3243x + 7.959, R^2 = 0.9976$) from 7.65 to 4.63 years when thermal efficiency increases from 45% to 73%. This is because higher thermal efficiency leads to increases in the cash flow and annual savings which consequently increases the NPV and decreases the payback period. In contrast, with an increase in thermal efficiency electricity generation increases, and solar cost decreases which in combined decreases the LCWH.

The results of the simulation reveal that a solar water heater with a solar collector of higher thermal efficiency should be preferred for the higher value of NPV and lower value of LCWH/PP.

4.3.7. Single-objective optimization

Three output parameters namely Levelized cost of water heating (LCWH), net present value (NPV) & payback period (PP) have been considered to be optimization objectives. On the other hand cost of collector (C_0), the debt-equity ratio (DE_r), interest rate (i_r), operational and maintenance costs (OM_N), inflation rate (if_r), solar irradiance (I_T), collector efficiency (η_{th}), electricity price (EP), and market discount rate (MD_r) have been considered to be decision variables.

In the EES software, several optimization methods have been presented. In this study, for optimization purposes, the Genetic Algorithm (GA) method has been considered. Many research studies have shown that the Genetic algorithm (GA) is more efficacious in comparison to the traditional mathematical & direct search methods [30,31]. To compute better approximations of the optimal solution, the GA method runs on a populace of potential solutions. GA method used in EES is procured from the PIKAIA optimization program written in FORTRAN 77. The genetic algorithm parameters for the optimization of objective functions are

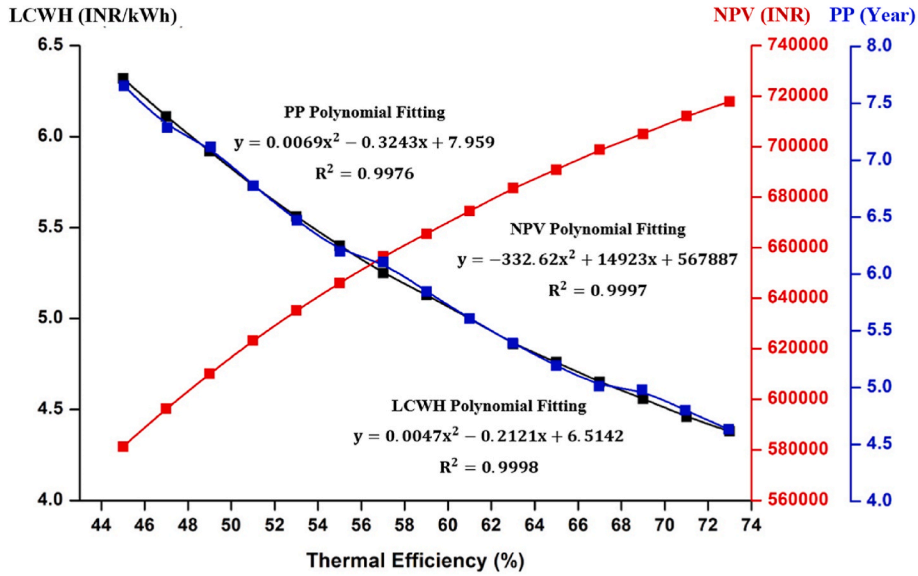


Fig. 16. Impact of variation of thermal efficiency on LCWH, NPV, and PP.

given in Table 4. The detailed procedure of the optimization is presented in Fig. 17.

The optimization is performed for the maximization of NPV and minimization of LCWH and payback period separately with the following considered constraints or restrictions as follows:

Optimize LCWH or NPV or PBP$\left(C_0, DE_r, i_r, OM_N, if_r, I_T, \eta_{th}, EP, MD_r \right)$

Subjected to

$30000 \le C_0 \le 35000$

$50 \le DE_r \le 90$

$8 \le i_r \le 10$

$1 \le OM_N \le 1.25$

$3.5 \le I_T \le 6.98$

$51 \le \eta_{th} \le 69$

$6 \le EP \le 7$

$4.5 \le MD_r \le 5.5$

Table 5 provides the optimal values of objective functions & decision variables, for comparison purposes, the base case or non-optimized values have also been presented in the same table. It is revealed that in respect of the base case, the value of LCWH was found to be lower by 33.46%, 25.34%, and 26.93% for LCWHOL, NPVOL, and PPOL cases respectively. Whereas NPV has increased by approximately 9%, 28.43%, and 26.35% for LCWHOL, NPVOL, and PPOL cases respectively. Furthermore, the value of the payback period was observed to be lowered by 37.76%, 41.43%, and 47.37% for LCWHOL, NPVOL, and PPOL cases respectively.

Table 4
Genetic Algorithm parameters.

S. No	Parameter	Value
1	No. of individuals in the model	9
2	Highest rate of mutation	0.25
3	No. of generations	64
4	Lowest rate of mutation	0.0005
5	Crossover rate (probability)	0.85
6	Initial rate of mutation	0.005

4.3.8. *Detailed output matrices of different zones*

Table 6 displayed the output parameters of the scenario model for five different zones according to the Indian market. Based on the input assumption, the mean value of LCWH, NPV, and PP are found to be 5.14 INR/kWh, 663788.48 INR, and 5.84 years respectively.

This table shows that the yearly energy demand to accomplish hot water demand is highest for Zone-I due to the lowest average inlet temperature of the water. Whereas due to the highest average inlet water temperature, the yearly energy demand for Zone-III is the lowest one.

It has been found that Zone-V would require the least collector area to meet the yearly energy demand for water heating whereas Zone-I required the largest collector area. This is because the collector area for a specific location is proportional to the ratio of yearly energy demand to the average yearly solar irradiance incident in the same location. This ratio was found to be the lowest for Zone-V due to the incident of the highest solar radiation and observed to be highest for Zone-I due to the highest yearly energy demand.

Furthermore, both LCWH and PP were found to be lowest and highest for Zone-V and Zone-III respectively. While NPV was observed to be highest for Zone-I and lowest for Zone-III. Because both LCWH & PP are indirectly proportional to solar radiation. The reason for the highest value of both LCWH and PP for Zone-III is because falling of the lowest average yearly solar irradiance.

However, due to the falling highest average yearly solar irradiance, the value of both LCWH and PP was found to be lowest for Zone-III.

On the other hand, NPV is the net present value of a system that depends on the total revenue generation at the end of its life. Since total revenue generation of a solar collector depends on the yearly energy demand which was found to be highest in Zone-I and lowest in Zone-III. Therefore, NPV was found to be highest for Zone-I and lowest for Zone-III.

5. Conclusions

In the present study, a computation model has been developed by the amalgamation of the Monte Carlo Technique (MCT) & Multi-energy/economic relations that take into consideration the risks and integrated uncertainties of different variables. The present work sheds light on the financial status of solar water heating technologies and provides

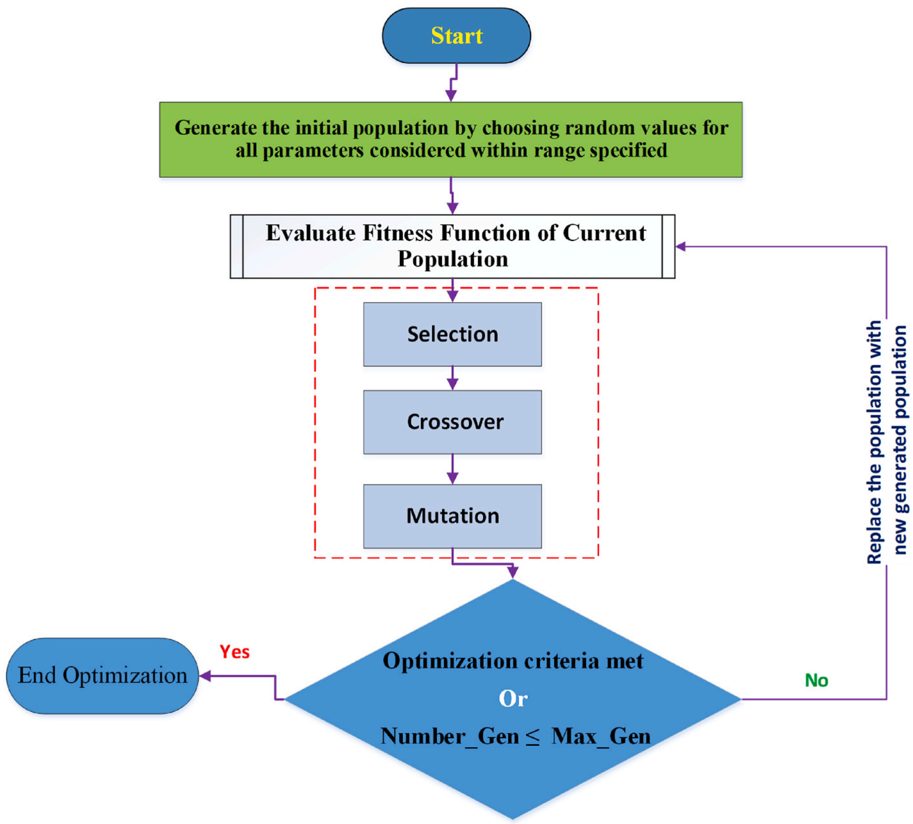


Fig. 17. Algorithm for optimization of output economic parameters.

Table 5
Comparison of optimized and non-optimized values for output matrices.

Decision Variables/ Objective Functions	Non-Optimized Value	Optimized Values			Unit
		LCWHOL	NPVOL	PPOL	
C_0	32,500	30,140	31,145	30,267	INR/m^2-Aperture area
DE_r	70	72.14	74.64	52.63	%
i_r	9	8.07	8.29	9.64	%
OM_N	1.25	1.02	1.02	1.22	%
if_r	4.5	4.42	4.56	4.21	%
I_T	5.24	6.98	6.80	6.72	kWh/m^2/day
η_{th}	60	68.72	66.79	69	%
EP	6.50	6.03	7	6.92	INR/kWh
MD_r	5	4.77	4.70	4.80	%
LCWH	5.05	3.36	3.77	3.69	INR/kWh
NPV	670,810	730,716	861,574	847,573	INR
PP	5.72	3.56	3.35	3.01	Year

facts for investment in India. Based on the simulation study, crucial conclusions have been drawn here as below:

❖It has been found that daily average solar radiation made the greatest impact on output economic matrices (LCWH, NPV, and PP). Therefore, higher daily average solar radiation will lead to better business results.
❖Based on the results, It is suggested that solar water heaters should be installed in a location where the daily average solar radiation exceeds 1 kWh/m2/day. Fortunately, all zones of India receive daily average solar radiation higher than 1 kWh/m2/day which is a good sign for the installation of the solar water heater.
❖Based on the input assumption, the mean value of LCWH, NPV, and PP are found to be 5.14 INR/kWh, 663788.48 INR, and 5.84 years respectively.
❖Zone-V would require the least collector area to meet the yearly energy demand for water heating up to desired temperature range whereas the largest area will be required for Zone-I.
❖Both LCWH and payback period were found to be lowest and highest for Zone-V and Zone-III respectively, While NPV was observed to be highest for Zone-I and lowest for Zone-III.
❖Among the selected cities, Ahmedabad and Srinagar achieve the highest (5.615 kWh/m2/day) and lowest (4.695 kWh/m2/day) daily average solar radiation respectively. It signifies that Zone-V is the most favorable place for the installation of the solar water heater.
❖Based on the optimization analysis, it is revealed that in respect of the base case, the value of LCWH was found to be lower by 33.46%, 25.34%, and 26.93% for LCWHOL, NPVOL, and PPOL cases respectively. Whereas NPV has increased by approximately 9%, 28.43%, and 26.35% for LCWHOL, NPVOL, and PPOL cases respectively. Furthermore, the value of the payback period was observed to be lowered by 37.76%, 41.43%, and 47.37% for LCWHOL, NPVOL, and PPOL cases respectively.
❖The Monte Carlo method considers the integration of risk and uncertainties of key input variables whose simulation outcomes would be more beneficial for realistic investment decisions on HPT-ETCS technologies in the Indian building sector.

It is expected that the above discussion boosts business tycoons to have more investment in solar water heaters. The installation of solar water heaters should be prioritized in those municipalities or regions of India where electricity prices are high. In addition to this, for healthier market penetration, the Indian government may provide a suitable subsidy on solar water heaters.

Table 6
Parameters of scenario model for different zones of India.

S. No	Parameters	Zone-I	Zone-II	Zone-III	Zone-IV	Zone-V
1	No. of Evacuated Tubes	65	45	47	50	42
2	Collector Area (m^2)	6.09	4.28	4.45	4.68	3.99
3	Capital Investment (INR)	197,925	139,100	144,625	152,100	129,675
4	Total Loan Cost (INR)	285209.96	200443.05	208404.71	219176.10	186861.71
5	Total O&M Cost (INR)	51421.01	36138.17	37573.66	39515.86	33689.48
6	Energy Generated by Solar System (kWh)	111431.86	81257.72	75793.45	84676.05	76934.77
7	Auxiliary Energy Cost (INR)	169618.33	122888.49	114243.64	129529.25	115033.49
8	Total Cost by Solar System (INR)	506249.30	359469.49	360222.01	388221.21	335584.68
9	Total Cost by Conventional System (INR)	1801439.49	1312835.95	1224171.7	1369535.18	1241675.32
10	Life Cycle Savings (INR)	1295190.19	953366.24	863949.69	981313.97	906090.64
11	LCWH (INR/kWh)	4.57	4.44	4.80	4.62	4.38
12	NPV (INR)	779965.47	575569.42	517497.43	590477.53	547584.45
13	PP (Year)	4.99	4.76	5.28	5.06	4.66

Credit author statement

K. Chopra: Conceptualization, Formal analysis, Writing – original draft, Supervision.V. V. Tyagi: Methodology, Data curation, Conceptualization, Supervision, Writing – review & editing Sakshi Popli: Data curation, Data Validation.A. K. Pandey: Validation, Writing – review & editing,

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Data availability

No data was used for the research described in the article.

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